

(Please write your Enrollment Number)

Enrollment No. 01701012022

MID-TERM EXAMINATION

(Course Name: B. Tech) (Semester: 6th)

(March 2025) OFF LINE mode

Subject Code: BIT 304	Subject: Cloud Computing
Time : 1 ½ Hours	Maximum Marks :30

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1		(2.5*4=10)	CO Mapping
	a) Explain how Fog Computing supports real-time data processing applications and how it differs from traditional cloud computing.		CO1
	b) According to NIST, what are the essential characteristics of cloud computing?		CO1
	c) Differentiate between resource virtualization and server virtualization using suitable real-world example.		CO2
	d) How does cloud-optimized storage improve performance, scalability, and cost-efficiency in a cloud-based infrastructure?		CO2
UNIT I			CO Mapping
Q2	Attempt any two parts	(5*2=10)	
	a) An enterprise is transitioning from a traditional IT model to a cloud-based SPI framework. Analyze the key differences between these two models and the potential advantages of SPI.		CO1
	b) A business needs to decide on a cloud deployment strategy. Discuss the various cloud deployment models and the factors influencing their selection.		CO1
	c) Explain how cloud computing supports green IT initiatives and contributes to energy efficiency with real world example.		CO1
UNIT II			CO Mapping
Q3	Attempt any two parts	(5*2=10)	
	a) Discuss the key components of storage network architecture and their role in efficient data handling.		CO2
	b) Discuss the role of Microsoft Hyper-V in server virtualization and its impact on enterprise computing.		CO2
	c) What is network virtualization? Explain its types and how it enhances network performance in cloud environments.		CO2

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MID-TERM EXAMINATION
(Course Name: B. Tech.) (Semester: 6)
(March 2025) OFF LINE mode

Subject Code: BCS 306	Subject: Compiler Design
Time : 1 ½ Hours	Maximum Marks :30

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1		(2.5*4=10)	CO Mapping
	a) What is CFG? In which phase of compiler CFG is used? Discuss about the difference between Leftmost Derivation and Rightmost Derivation with suitable example.		CO ₁
	b) With a neat diagram discuss about bootstrapping and its uses.		CO ₁
	c) What are the rules useful to find FIRST and FOLLOW, explain with suitable example.		CO ₂
	d) What is parsing? Briefly discuss about different types of parsing techniques.		CO ₁
UNIT I			CO Mapping
Q2	Attempt any two parts	(5*2=10)	
	a) Discuss about different Phases of a compiler with a neat diagram.		CO ₁
	b) What is the role of Lexical Analyzer? Discuss about Input Buffering and its uses.		CO ₁
	c) Define Ambiguity. Check whether the following grammars are ambiguous or not. Grammar 1: $A \rightarrow AA/(A)/a$ Grammar 2: $E \rightarrow EE/AB$ $A \rightarrow Aa/a$ $B \rightarrow Bb/b$		CO ₂
UNIT II			CO Mapping
Q3	Attempt any two parts	(5*2=10)	
	a) Discuss about Recursive Descent Parser (RDP) and design RDP for the following grammar. $S \rightarrow AA$ $A \rightarrow aA$ $A \rightarrow b$		CO ₁ & CO ₂
	b) Discuss about closure and goto operations with suitable example, and Construct LR(0) items sets for the following grammar. $A \rightarrow BB$ $B \rightarrow cB/d$		CO ₂
	c) Construct the SLR parse table for the following grammar. $S \rightarrow cAd$ $A \rightarrow ab/e$ And parse the string: cabd		CO ₃

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Enrollment No. 01701012022

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name: B.Tech. Semester: 6th
(May, 2025)

Subject Code: BCS 306	Subject: Compiler Design
Time :3 Hours	Maximum Marks :60

Note: Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		(2.5*8=20)	CO Mapping
	a) What is the purpose of symbol table in compilation? How is it used during lexical analysis?	2.5	CO1
	b) Differentiate between Syntax analysis and Semantic analysis.	2.5	CO1
	c) Explain Predictive parsing.	2.5	CO2
	d) Discuss about different actions possible in Shift Reduce parsing using suitable example.	2.5	CO2
	e) What is a type system? Give two examples of static type checking.	2.5	CO3
	f) Suppose $\Sigma=\{a,b\}$, then what type strings will appear in $L(a^*b^*)$.	2.5	CO2
	g) Explain about Abstract Syntax Tree and Directed Acyclic Graph using suitable example.	2.5	CO4
	h) Convert the following expression in to equivalent three address code. $(a+b*c)+d+(a+b*c)-d+e$	2.5	CO4
UNIT I			CO Mapping
Q2	Explain the concept of "Grouping of Phases" in compiler design. Discuss how front-end and back-end phases are grouped, along with their significance in compiler construction.	(10)	CO1
Q3	Define regular expression. Let $\Sigma=\{0,1\}$ then what type of language does regular expression $(0+1)^*$ will generate Construct a deterministic finite Automate for $(0+1)^*$.	(10)	CO2
UNIT II			CO Mapping
Q4	Given the grammar: $S \rightarrow CC$ $C \rightarrow cC d$ Construct the Canonical LR (CLR) parsing table and parse the input "ccdcd".	(10)	CO4
Q5	Given the augmented grammar: $S' \rightarrow S$ $S \rightarrow L=R R$ $L \rightarrow *R id$ $R \rightarrow L$ a) Construct the canonical LR(1) item sets b) Construct the LALR(1) parsing table	(10)	CO4
UNIT III			CO Mapping
Q6	Discuss about different types of attributes? And compare S-attributed and L-attributed grammars with suitable examples.	(10)	CO3
Q7	How static storage allocation is different from stack-based storage allocation explain with neat diagrams.	(10)	CO3

UNIT IV

Q8	Explain the following optimizations using suitable examples: • Common sub expression elimination • Reduction in strength • Constant folding • Code motion • Dead code elimination	(10)	CO Mapping
Q9	What are Lexical phase errors, syntactic errors and semantic errors? Explain with example. How these errors are detected.	(10)	CO2

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Enrollment No. 01701012022

MID-TERM EXAMINATION

(B. Tech) (Semester:6th)

(March 2025) OFF LINE mode

Subject Code: BCS 304	Subject: Microprocessor and Interfacing
Time : 1 ½ Hours	Maximum Marks :30

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1		(2.5*4=10)	CO Mapping
	a) Do 8086 implements pipelining? Justify your statement		CO1
	b) Can we change the content of the Code Segment directly? Justify your statement.		CO1
	c) Differentiate and explain the following instructions: 1. SUB CL,BL 2. SUB CX,AX 3. SUB DX,0016H 4. SUB CX,[DI] 5. SUB [BP] AH		CO2
	d) Given AX=0102H, BX=0304H, SP=0722, and SS=0200. After executing the following program what will be the contents of AX, BX, SP, and STACK : POP BX POB AX		CO2
UNIT I			CO Mapping
Q2	Attempt any two parts	(5*2=10)	
	a) Explain the function of the following pins in 8085: INTR, ALE, READY, HOLD, TRAP		CO1
	b) Explain in detail the demultiplexing of the Address bus in 8086		CO1
	c) Explain the function of EU, BIU, and QUEUE in 8086		CO1
UNIT II			CO Mapping
Q3	Attempt any two parts	(5*2=10)	
	a) IF SS=0100H, DS=2000H, ES=1002, BX=1001H, BP=2100, SI=0002H. Mention the contents of BX, BP, SI, ES, and SS after the execution of LES BX, [BP]+[SI]		CO2
	b) Draw the timing diagram of the write cycle in 8086 Minimum mode		CO2
	c) Explain the following assembler directives: ASSUME, ENDP, EXTRN, ORG, ENDS		CO2

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End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name:<B.Tech CSE>, Semester:<6th>
(May, 2025)

Subject Code: BCS 304	Subject: Microprocessor and Interfacing
Time :3 Hours	Maximum Marks :60
Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.	

Q1		(2.5*8=20)	CO Mapping
	(a) What are Coprocessors why they are needed? Give examples also.		CO1
	(b) How the low byte flag register can be modified in 8086?		CO1
	(c) Distinguish between CALL and JUMP.		CO2
	(d) Draw timing diagram of IO write cycle in 8086.		CO2
	(e) Explain the significance of the use of 8259.		CO3
	(f) Differentiate Edge triggered and Level triggered Interrupts with examples.		CO3
	(g) Explain the need of Framing. How it is done?		CO4
	(h) Justify the need of Interfacing.		CO 4
UNIT-I			CO Mapping
Q2	a) What is meant by microprocessor architecture and explain its operation. If a microprocessor is of 32 bit size what does it means, explain in detail. b) Discuss the function of Queue. How the concept of pipelining is implemented in 8086.	(5,5)	CO1
Q3	a) What is Segmentation. Explain the advantage of Segmentation Scheme in 8086. Write the size of physical memory and virtual memory of 8086 microprocessor. b) Explain the function of the following pins in 8086: $\overline{DT/R}$, $\overline{DEN}$, ALE , $\overline{TEST}$, $READY$, $\overline{BHE}$	(5,5)	CO1
UNIT-II			CO Mapping
Q4	a) Write short note on: (i) Loop, (ii) Flag manipulation, (iii) shift & rotate. b) What is physical address in 8086. Assume the following state of 8086 microprocessor: $CS = 1000H$, $DS = 2000H$, $SS = 3000H$, $ES = 4000H$, $BP = 0010H$ $BX = 0020H$, $SP = 0030H$, $SI = 0040H$, $DI = 0050H$. Calculate physical address for following instructions: i) $MOV CX, [BX]$, ii) $MOV AL, [BP+SI]$, iii) $MOV CS:[BX], AL$, iv) $MOV AL, [BP]$.	(3+7)	CO2
Q5	a) Write a program to add a data byte located at offset 0500H in 2000H segment to another data byte available at 0600H in the same segment and store the result at 0700H in the same segment. b) What is the purpose of the assembler in the 8086 microprocessor? Differentiate between assembler directive and instructions in 8086. Explain any three assembler directives.	(5,5)	CO2
UNIT-III			CO Mapping
Q6	a) What is Multiprocessing systems? Compare Symmetrical and Asymmetrical Multiprocessing systems. b) Compare the features of 80386 and 80286 based on the following points: registers, clock rate, number of pins, number of instructions, modes of operations, memory management, and protection mechanism.	(5,5)	CO3

Q7	a) What does the CPU do when it receives an interrupt? How do you enable and disable interrupts in 8086? b) Draw and Explain the Interfacing of 8259 with 8086. What is the advantage of using 8259.	(5,5)	CO3
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UNIT-IV

CO Mapping

Q8	a) Write the sequence of instructions required to initialize 8251A at address 80H and 81H for the configuration given below. (i) character length-6 bits (iv) Baud rate 64 X (it) Parity even (v) DTR and RTS asserted (iiz) Stop bit 1 (vi) Error flag reset (vit) Transmitter enable b) What are the functions of DMA controller? Describe the steps that take place during DMA operation.	(5,5)	CO4
Q9	a) Discuss the process of interfacing with the 8255 programmable peripheral interface using the 8086 microprocessor, including the various modes of operation. b) What is DMA, why it is required. Explain Cycle stealing and Burst mode.	(5,5)	CO4

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Enrollment No. 01701012022

MID-TERM EXAMINATION
(Course Name: B. Tech) (Semester: 6 Sem Jan-May 2025)
(March 2025) OFF LINE mode

Subject Code: BCS 310	Subject: Human-Computer Interaction
Time : 1 ½ Hours	Maximum Marks :30

Note: Q1 is compulsory. Attempt any twoparts of Q2 and Q3

Q1		(2.5*4=10)	CO Mapping
	a) Define the term "User Interface" and explain its importance in software applications.		CO1
	b) Discuss the key principles of an effective Graphical User Interface (GUI) design.		CO1
	c) Explain the significance of human characteristics in the design process of Human-Computer Interaction.		CO2
	d) Discuss the key constraints and limitations of mobile and hand-held platforms compared to other computing platforms.		CO2
UNIT I			
Q2	Attempt any two parts	(5*2=10)	CO Mapping
	a) Explain the evolution of screen design and how it has influenced modern user interface development.		CO1
	b) Analyze the characteristics of web user interfaces and how they differ from other types of user interfaces.		CO1
	c) Illustrate the role of direct manipulation in graphical systems and discuss its impact on user experience with proper examples.		CO1
UNIT II			
Q3	Attempt any two parts	(5*2=10)	CO Mapping
	a) How does human interaction speed affect the design of user interfaces? Provide relevant examples.		CO2
	b) Evaluate the importance of understanding business functions while designing user interfaces for enterprise applications.		CO2
	c) Propose innovative solutions to improve UI responsiveness for mobile and handheld platforms under constraints like network, power, and memory.		CO2

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Enrollment No. 01701012022

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(Offline)
Course Name: B.Tech., Semester:6
(May, 2025)

Subject Code: BCS 310	Subject: Human Computer Interaction
Time :3 Hours	Maximum Marks :60

Note:Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1	a) Define "User Interface" and explain its significance in modern computing. b) Differentiate between Hover-Reveal Tools and Toggle-Reveal Tools. c) What are the challenges present in display based Systems. d) Explain the principle of "direct manipulation" in the context of graphical user interfaces. e) Discuss the importance of screen design in user interfaces. f) What are the common limitations of mobile and hand-held platforms in terms of user interface design? g) What is the role of colors and multimedia in enhancing user experience in UI design? h) How does human interaction speed influence user interface design and usability?	(2.5*8=20)
UNIT I		
Q2	Justify the statement with example: "The Response Time/Productivity Relation changes based on the task and user."	(10)
Q3	Describe the principles of user interface design for graphical user interfaces (GUIs). Explain the concept of direct manipulation and how it is used in modern graphical systems.	(10)
UNIT II		
Q4	Explain the design process for user interfaces. Discuss the role of human interaction characteristics (such as cognitive factors and physical limitations) in shaping the design of user interfaces.	(10)
Q5	Explain the following Elements of Mobile Design in detail a. Context b. Message c. Look and Feel d. Layout e. Color	(10)
UNIT III		
Q6	Discuss the following characteristics in respect of web pages a. Page size b. Page rendering c. Page layout d. Page resolution e. User focus	(10)
Q7	Discuss the importance of navigation schemes in user interface design. How do you select the appropriate window types and controls for an application? Provide examples.	(10)
UNIT IV		
Q8	Explain the role of multimedia, icons, and colors in user interface design. How can these elements be utilized to enhance the user experience and engagement?	(10)
Q9	Discuss the various interacting devices used in user interfaces, such as keyboards, touchscreens, and voice-activated commands. How do these devices impact user interaction and usability?	(10)

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MID-TERM EXAMINATION
(Course Name: B. Tech.) (Semester: 6th)
(March 2025) OFF LINE mode

Subject Code: HMC-306	Subject: Financial Management
Time : 1 ½ Hours	Maximum Marks :30

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1															
	a) What is Financial Management, and why is it important for businesses?		(2.5*4=10)												
	b) Explain the Goal of Financial Management.														
	c) Explain the difference between bonds and debentures.														
	d) Explain the factors affecting capital structure.														
UNIT I															
Q2	Attempt any two parts														
	a) Define the Time Value of Money (TVM). Explain its significance with suitable examples and outline its key assumptions.		(5*2=10)												
	b) Describe the type of Decisions a financial manager has to take. Explain how these decisions are interrelated.														
	c) Mr Dhawan borrowed Rs. 7, 00,000 to buy a Machine. If he pays Equal Installments for 20 Years @8% P.A. Compute Amount of Installments														
UNIT II															
Q3	Attempt any two parts														
	a) 1. What do you understand by the term perpetual and irredeemable preference share?		(5*2=10)												
	2. Define cost of Equity with relevant examples?	2.5													
		2.5													
	b) Define WACC? Explain its significance in Financial decision-making with relevant examples.		5												
	c) The following is the capital structure of ABC Ltd.		5												
	<table><tr><th>Sources of Capital</th><th>Amount</th><th>Cost of Capital of Sources</th></tr><tr><td>Equity Share Capital (shares of Rs. 500 each)</td><td>60,00,000</td><td>18%</td></tr><tr><td>14% Preference Share Capital (shares of Rs. each)</td><td>10,00,000</td><td>?</td></tr><tr><td>12% Debentures of Rs. Each</td><td>30,00,000</td><td>?</td></tr></table>	Sources of Capital	Amount	Cost of Capital of Sources	Equity Share Capital (shares of Rs. 500 each)	60,00,000	18%	14% Preference Share Capital (shares of Rs. each)	10,00,000	?	12% Debentures of Rs. Each	30,00,000	?		
Sources of Capital	Amount	Cost of Capital of Sources													
Equity Share Capital (shares of Rs. 500 each)	60,00,000	18%													
14% Preference Share Capital (shares of Rs. each)	10,00,000	?													
12% Debentures of Rs. Each	30,00,000	?													
The tax rate for the given period is 20%. Calculate WACC.															

Calculators are allowed.

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name: B.Tech, Semester: 6th
(December, 2024)

Subject Code: HMC 306	Subject: Financial Management
Time :3 Hours	Maximum Marks :60

Note: Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Compulsory. Attempt one question each from the Units I, II, III & IV.											
Q1		(2.5*8=20)									
	a) What do you understand by the rule 72? Explain with examples										
	b) Explain the term financial management?										
	c) Explain the concept of the Interest Tax Shield?										
	d) What do you understand by the term Equity and how it is different from preference share?										
	e) What do you understand by the term capital budgeting, and to which type of decision does it belong?										
	f) You are given the present values of cash inflows and outflows for two projects (A and B). Calculate the Profitability Index (PI) for each project and determine which project is the most suitable for investment.										
	<table><tr><td></td><td>A</td><td>B</td></tr><tr><td>Present Value of Cash Outflows</td><td>75000</td><td>68500</td></tr><tr><td>Present Value of Cash Inflows</td><td>85500</td><td>74500</td></tr></table>		A	B	Present Value of Cash Outflows	75000	68500	Present Value of Cash Inflows	85500	74500	
	A	B									
Present Value of Cash Outflows	75000	68500									
Present Value of Cash Inflows	85500	74500									
	g) Explain storage and ordering cost in inventory management										
	h) Explain the concept of gross and net working capital.										
UNIT I											
Q2	A. Why is the time value of money important to consider? Explain it with proper examples	(5)									
	B. Jaya retires at the age of 60 years receiving pension of 10000 a year paid in yearly installments. If he expects his life for another 15 years, what single sum is equivalent to her pension (Interest rate = 6% p.a.)	(5)									
Q3	A. Define the scope of Financial Management. What role should the financial manager play in a modern enterprise?	(5)									
	B. Explain the factors affecting time value of money?	(5)									
UNIT II											
Q4	A. What is meant by the term cost of capital? What are the main components involved in calculating it? Explain each component with suitable examples.	(5) (5)									
Q5	XYZ Ltd has the following capital structure <table><tr><td>Equity Share Capital (5,000 Shares)</td><td>30,00,000</td></tr><tr><td>9% Preference Shares (100 Rs. Each)</td><td>22,00,000</td></tr><tr><td>12% Debentures</td><td>28,00,000</td></tr></table> - The cost of Equity is 15% - The preference shares are issued at a floatation cost of 2% and discount at 3%. Calculate cost of preference shares if these shares are irredeemable. - Debentures are issued at premium of 5%.	Equity Share Capital (5,000 Shares)	30,00,000	9% Preference Shares (100 Rs. Each)	22,00,000	12% Debentures	28,00,000	(10)			
Equity Share Capital (5,000 Shares)	30,00,000										
9% Preference Shares (100 Rs. Each)	22,00,000										
12% Debentures	28,00,000										

- Tax rate is given @20%.
1. Compute the cost of capital for each source
 2. Compute the Weighted Average Cost of Capital

UNIT III

- Q6** ABC Ltd. is evaluating two alternatives Machine X and Machine Y. The initial costs of Machine P and Machine Q are ₹25,00,000 and ₹22,50,000, respectively. The expected net cash flows for both machines over their useful lives are as follows:

Year	Machine P	Machine Q
1	8,50,000	5,85,000
2	7,55,000	3,83,000
3	4,95,000	7,50,000
4	3,50,000	6,50,000
5	7,44,000	4,38,000
6	4,50,000	6,80,000

The company has target return on capital from Machine X and Y are 12%
On the basis above information you are required to calculate the Net Present Value of both the machines and compare the profitability of the machines and state which alternative you consider to be financially preferable

- Q7** Write Short Notes with suitable examples on
- Net Present Value
 - Internal Rate of Return
 - Profitability Index
 - Payback Period

UNIT IV

- Q8** A. Explain ABC analysis model of Inventory Management with suitable examples (5)
- B. What is the operating cycle? Explain its concept and the method of calculation. (5)

- Q9** ABC Hospital Supplies Pvt. Ltd. purchases 9,000 units of surgical gloves annually, ordering one month's requirement at a time. Each pair of gloves costs ₹20. The ordering cost per order is ₹15, and the carrying cost is 15% of the unit cost. (10)
- A. Calculate the Economic Order Quantity (EOQ) and the Total Cost at the EOQ level.
- B. Calculate the Total Cost if the order sizes are:
- i. 100 units
 - ii. 600 units
 - iii. 3000 units
- C. Compare each of the above order sizes with the EOQ and comment on which ordering policy is the most cost-effective

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Enrollment No. 01701012022

(1)

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(Offline)
Course Name: B.Tech, Semester: 6
(May, 2025)

Subject Code: BCS 302	Subject: Wireless Networks
Time: 3 Hours	Maximum Marks :60

Note: Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		(2.5*8=20)	CO Mapping
a)	Differentiate between IEEE 802.11 and IEEE 802.15 standards.	2.5	CO1
b)	What is the role of multicast routing in ad-hoc networks, and provide one routing protocol that support multicast routing?	2.5	CO1
c)	Provide any two issues in designing the transport layer for ad-hoc networks according to infrastructure less setup.	2.5	CO2
d)	Provide the name of two simulators available for Mesh, adhoc and Wireless sensor networks, and a detailed comparison between them.	2.5	CO2
e)	Define Opportunistic Routing with an example.	2.5	CO3
f)	How are Vehicular Mesh Networks different from traditional mesh networks?	2.5	CO3
g)	Define data dissemination and data gathering in WSNs.	2.5	CO4
h)	List any two energy-aware routing protocols used in ad-hoc networks.	2.5	CO4
UNIT I			CO Mapping
Q2	Define ad-hoc networks and mention two of their key characteristics compare to traditional networks. Also provide a diagram of layer wise architecture of adhoc networks.	(10)	CO1
Q3	Describe the design goals and issues in MAC protocol design for ad-hoc networks. Elaborate on contention-based MAC protocols and their working principles.	(10)	CO1
UNIT 2			CO Mapping
Q4	There are 10 end devices in adhoc network. Each node has atleast two immediate neighbors. Find and demonstrate the complete process for the route from source to destination using dynamic source routing (DSR).	(10)	CO2
Q5	Describe hybrid routing protocols. How do they combine the benefits of both proactive and reactive approaches?	(10)	CO2
UNIT 3			CO Mapping
Q6	Explain the concept of heterogeneous mesh networks and their use cases in VANETS . How heterogenous networks help in building reliable transportation system compared with traditional wireless networks.	(10)	CO3
Q7	Consider a network scenario for Mesh networks that contain 20 wireless nodes connected using MESH topology. Client needs to setup a communication between X (Sender) to Y (Receiver). Demonstrate the results after applying the proactive, reactive, and geographic routing to the given scenario.	(10)	CO3
UNIT 4			CO Mapping
Q8	Why is energy efficiency critical in WSNs? Discuss and compare different approaches to achieve energy efficiency in WSNs.	(10)	CO4
Q9	Describe various location discovery techniques in wireless sensor networks. Consider a network scenario of sparse network with best location discovery approach to demonstrate the working with diagrammatical presentation.	(10)	CO4

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Enrollment No. 01701012022

MID-TERM EXAMINATION

(Course Name: B. Tech.) (Semester: 6th)

(March 2025) OFF LINE mode

Subject Code: BCS 302	Subject: Wireless Networks
Time : 1 ½ Hours	Maximum Marks :30

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1		(2.5*4=10)	CO Mapping
	a) List the key characteristics of wireless channels.	2.5	CO1
	b) Discuss IEEE 802.15 standard and its working groups.	2.5	CO1
	c) Describe in brief the path finding process in Cluster Head Gateway Switch routing protocol.	2.5	CO2
	d) Discuss the types of routing protocols based on routing information update mechanism.	2.5	CO2
UNIT I			CO Mapping
Q2	Attempt any two parts	(5*2=10)	
	a) Discuss the key features of Ad-hoc networks and how they differ from traditional wireless networks.	5	CO1
	b) A university wants to implement a wireless network for its campus. The network must support both indoor and outdoor connectivity. Design a solution and briefly discuss the challenges.	5	CO1
	c) Describe in detail RI-BTMA Protocol.	5	CO1
UNIT II			CO Mapping
Q3	Attempt any two parts	(5*2=10)	
	a) How does Location-Aided Routing (LAR) improve efficiency over traditional flooding?	5	CO2
	b) Analyze the challenges in designing a routing protocol for Ad-hoc wireless network.	5	CO2
	c) Describe in detail any one hybrid routing protocol.	5	CO2